

DATABASE MANAGEMENT SYSTEM

PRACTICAL FILE

RAMANUJAN COLLEGE



UNIVERSITY OF DELHI

SUBMITTED BY :

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EXAMINATION ROLL NO :

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CLASS : BSC(H) Computer Science
SEM-4

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```
mysql> show databases ;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.01 sec)

mysql> create database aditi_dbms
-> ;
Query OK, 1 row affected (0.01 sec)

mysql> show databases ;
+-----+
| Database |
+-----+
| aditi_dbms |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> use aditi_dbms;
Database changed
```

Creating table – student , society , enrolment:

```
mysql> create table student (
-> Roll_no char(6),
-> Student_name varchar(20),
-> Course varchar(10),
-> DOB DATE,
-> PRIMARY KEY(Roll_no)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> show tables;
+-----+
| Tables_in_aditi_dbms |
+-----+
| student |
+-----+
1 row in set (0.01 sec)
```

```
mysql> create table society(
-> soc_id char(6),
-> soc_name varchar(20),
-> mentor_name varchar(15),
-> totalseats int UNSIGNED );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> show tables ;
+-----+
| Tables_in_aditi_dbms |
+-----+
| society |
| student |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> alter table society
-> add primary key (soc_id);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> create table enrollment (
-> Roll_no char(6),
-> SID char(6),
-> date_of_enrollment DATE );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> alter table enrollment
-> add foreign key(Roll_no) references student(Roll_no);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> alter table enrollment
-> add foreign key (SID) references society(soc_id);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> show tables ;
+-----+
| Tables_in_aditi_dbms |
+-----+
| enrollment            |
| society               |
| student               |
+-----+
```

Inserting a sample data into all the tables :

```
mysql> INSERT INTO STUDENT (Roll_no, Student_name, Course, DOB) VALUES
-> ('X12349', 'Alice', 'comp sci', '2003-05-21'),
-> ('Y56781', 'Bob', 'chemistry', '2002-08-15'),
-> ('Z98769', 'Charlie', 'physics', '2001-12-10'),
-> ('X99939', 'David', 'maths', '2004-01-09'),
-> ('A12345', 'Annie', 'comp sci', '2003-03-30'),
-> ('B67890', 'Eve', 'biology', '2002-11-25'),
-> ('C11111', 'Frank', 'chemistry', '2001-07-17');
Query OK, 7 rows affected (0.01 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO SOCIETY (soc_id, soc_name, mentor_name, totalseats) VALUES
-> ('S001', 'NSS', 'Dr.Smith', 50),
-> ('S002', 'Robotics Club', 'Ms.Johnson', 30),
-> ('S003', 'Drama Club', 'Mr.Lee', 40),
-> ('S004', 'Science Society', 'Dr.Watson', 60),
-> ('S005', 'Art Society', 'Ms.Patel', 25);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO ENROLLMENT (Roll_no, SID, date_of_enrollment) VALUES
-> ('X12349', 'S001', '2023-08-01'),
-> ('Y56781', 'S002', '2023-08-02'),
-> ('Z98769', 'S001', '2023-08-03'),
-> ('X99939', 'S003', '2023-08-04'),
-> ('A12345', 'S004', '2023-08-05'),
-> ('A12345', 'S001', '2023-08-06'),
-> ('B67890', 'S002', '2023-08-07'),
-> ('C11111', 'S005', '2023-08-08');
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

```
MySQL 8.0 Command Line Cli X + v
mysql> select * from student;
+-----+-----+-----+-----+
| Roll_no | Student_name | Course | DOB |
+-----+-----+-----+-----+
| A12345 | Annie | comp sci | 2003-03-30 |
| B67890 | Eve | biology | 2002-11-25 |
| C11111 | Frank | chemistry | 2001-07-17 |
| X12349 | Alice | comp sci | 2003-05-21 |
| X99939 | David | maths | 2004-01-09 |
| Y56781 | Bob | chemistry | 2002-08-15 |
| Z98769 | Charlie | physics | 2001-12-10 |
+-----+-----+-----+-----+
7 rows in set (0.04 sec)

mysql> select * from society;
+-----+-----+-----+-----+
| soc_id | soc_name | mentor_name | totalseats |
+-----+-----+-----+-----+
| S001 | NSS | Dr.Smith | 50 |
| S002 | Robotics Club | Ms.Johnson | 30 |
| S003 | Drama Club | Mr.Lee | 40 |
| S004 | Science Society | Dr.Watson | 60 |
| S005 | Art Society | Ms.Patel | 25 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from enrollment ;
+-----+-----+-----+
| Roll_no | SID | date_of_enrollment |
+-----+-----+-----+
| X12349 | S001 | 2023-08-01 |
| Y56781 | S002 | 2023-08-02 |
| Z98769 | S001 | 2023-08-03 |
| X99939 | S003 | 2023-08-04 |
| A12345 | S004 | 2023-08-05 |
| A12345 | S001 | 2023-08-06 |
| B67890 | S002 | 2023-08-07 |
| C11111 | S005 | 2023-08-08 |
+-----+-----+-----+
8 rows in set (0.00 sec)

mysql> select Student_name from student
```

Practical questions:

1. Retrieve names of students enrolled in any society.
 2. Retrieve all society names.
 3. Retrieve students' names starting with the letter 'A'.
 4. Retrieve students' details studying in courses 'computer science' or 'chemistry'.
-
5. Retrieve students' names whose roll no either starts with 'X' or 'Z' and ends with '9'
 6. Find society details with more than N TotalSeats where N is to be input by the user
 7. Update society table for the mentor name of a specific society
 8. Find society names in which more than five students have enrolled
 9. Find the name of the youngest student enrolled in society 'NSS'
 10. Find the name of the most popular society (on the basis of enrolled students)

```
mysql> select Student_name from student
      -> join enrollment on student.Roll_no=enrollment.Roll_no;
```

Student_name
Annie
Annie
Eve
Frank
Alice
David
Bob
Charlie

8 rows in set (0.01 sec)

```
mysql> select soc_name from society;
```

soc_name
NSS
Robotics Club
Drama Club
Science Society
Art Society

5 rows in set (0.00 sec)

```
mysql> select Student_name from student
      -> where Student_name like 'A%';
```

Student_name
Annie
Alice

2 rows in set (0.01 sec)

```
mysql> select * from student
      -> where Course in ('comp sci','chemistry');
```

Roll_no	Student_name	Course	DOB
A12345	Annie	comp sci	2003-03-30
C11111	Frank	chemistry	2001-07-17
X12349	Alice	comp sci	2003-05-21
Y56781	Bob	chemistry	2002-08-15

4 rows in set (0.00 sec)

```
mysql> select Student_name from student
      -> where (Roll_no like 'X%' or Roll_no like 'Z%') and Roll_no like '%9';
```

Student_name
Alice
David
Charlie

3 rows in set (0.01 sec)

```
mysql> select * from society;
```

soc_id	soc_name	mentor_name	totalseats
S001	NSS	Dr.Smith	50
S002	Robotics Club	Ms.Johnson	30
S003	Drama Club	Mr.Lee	40
S004	Science Society	Dr.Watson	60
S005	Art Society	Ms.Patel	25

```
5 rows in set (0.00 sec)
```

```
mysql> update society
```

```
    -> set mentor_name = 'Mrs.ADITI'
```

```
    -> where soc_name = 'Art Society';
```

```
Query OK, 1 row affected (0.03 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from society
```

```
    -> ;
```

soc_id	soc_name	mentor_name	totalseats
S001	NSS	Dr.Smith	50
S002	Robotics Club	Ms.Johnson	30
S003	Drama Club	Mr.Lee	40
S004	Science Society	Dr.Watson	60
S005	Art Society	Mrs.ADITI	25

```
5 rows in set (0.00 sec)
```

```
mysql> select soc_name from society
```

```
    -> join enrollment on society.soc_id=enrollment.SID
```

```
    -> group by society.soc_name
```

```
    -> having count(enrollment.Roll_no)>2;
```

soc_name
NSS
Robotics Club

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT s.Student_name
```

```
    -> FROM student s
```

```
    -> JOIN enrollment e ON s.Roll_no = e.Roll_no
```

```
    -> JOIN society so ON e.SID = so.soc_ID
```

```
    -> WHERE so.soc_name = 'NSS'
```

```
    -> ORDER BY s.DOB DESC
```

```
    -> LIMIT 1;
```

Student_name
Alice

```
1 row in set (0.00 sec)
```

```
mysql> select s.soc_name from society s
-> join enrollment e
-> on e.SID=s.soc_ID
-> group by soc_ID
-> order by count(e.Roll_no) desc
-> limit 1;
```

soc_name
NSS

```
mysql> select s.soc_name from society s
-> join enrollment e
-> on e.SID=s.soc_ID
-> group by soc_name
-> order by count(e.Roll_no)
-> limit 2;
```

soc_name
Art Society
Science Society

2 rows in set (0.00 sec)

```
mysql> select s.Student_name from student s
-> join enrollment e
-> on e.Roll_no = s.Roll_no
-> where e.SID is NULL ;
```

Student_name
Aditi
Shreya

2 rows in set (0.00 sec)

```
mysql> select * from enrollment;
```

Roll_no	SID	date_of_enrollment
X12349	S001	2023-08-01
Y56781	S002	2023-08-02
Z98769	S001	2023-08-03
X99939	S003	2023-08-04
A12345	S004	2023-08-05
A12345	S001	2023-08-06
B67890	S002	2023-08-07
C11111	S005	2023-08-08
C11111	S002	2023-06-05
A12234	NULL	NULL
D12345	NULL	NULL

11 rows in set (0.00 sec)

```
mysql> select S.Student_name from student s
-> join enrollment e
-> on e.Roll_no =s.Roll_no
-> group by s.Roll_no
-> having count(e.Roll_no)>=2;
```

```
+-----+
| Student_name |
+-----+
| Annie        |
| Frank        |
+-----+
```

2 rows in set (0.00 sec)

```
mysql> use aditi_dbms;
Database changed
mysql> select s.soc_name from society s
-> join enrollment e
-> on e.SID = s.soc_id
-> group by soc_id
-> order by count(e.Roll_no) desc
-> limit 1;
```

```
+-----+
| soc_name |
+-----+
| NSS      |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select soc_name from society
-> where mentor_name like '%Lee%';
```

```
+-----+
| soc_name |
+-----+
| Drama Club |
+-----+
```

1 row in set (0.01 sec)

```
mysql> select soc_name from society so
-> join enrollment e on e.SID =so.soc_id
-> group by so.soc_id,so.totalseats
-> having count(e.Roll_no) = 0.1 * (so.totalseats) ;
```

```
+-----+
| soc_name |
+-----+
| Robotics Club |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select s.Student_name from student s
-> join enrollment e
-> on e.Roll_no=s.Roll_no
-> join society so on so.soc_id = e.SID
-> where e.SID is not null;
```

Student_name
Annie
Annie
Eve
Frank
Frank
Alice
David
Bob
Charlie

9 rows in set (0.02 sec)

```
mysql> select s.Student_name from student s
-> join enrollment e
-> on e.Roll_no=s.Roll_no
-> join society so on so.soc_id = e.SID
-> where so.soc_name in ('NSS','Art Society');
```

Student_name
Alice
Charlie
Annie
Frank

```
mysql> select s.soc_name ,s.totalseats-count(e.Roll_no) 'vacant seats' from society s
-> join enrollment e on e.SID= s.soc_id
-> group by s.totalseats, soc_id;
```

soc_name	vacant seats
NSS	47
Robotics Club	27
Drama Club	39
Science Society	59
Art Society	24

```
mysql> select * from society;
```

soc_id	soc_name	mentor_name	totalseats
S001	NSS	Dr.Smith	50
S002	Robotics Club	Ms.Johnson	30
S003	Drama Club	Mr.Lee	40
S004	Science Society	Dr.Watson	60
S005	Art Society	Mrs.ADITI	25

```
mysql> update society
-> set totalseats = totalseats + (0.1 * totalseats);
Query OK, 5 rows affected (0.01 sec)
Rows matched: 5  Changed: 5  Warnings: 0
```

```
mysql> select * from society
-> ;
```

soc_id	soc_name	mentor_name	totalseats
S001	NSS	Dr.Smith	55
S002	Robotics Club	Ms.Johnson	33
S003	Drama Club	Mr.Lee	44
S004	Science Society	Dr.Watson	66
S005	Art Society	Mrs.ADITI	28

```
mysql> alter table enrollment
-> add column fee_paid varchar (5);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> select * from enrollment;
```

Roll_no	SID	date_of_enrollment	fee_paid
X12349	S001	2023-08-01	NULL
Y56781	S002	2023-08-02	NULL
Z98769	S001	2023-08-03	NULL
X99939	S003	2023-08-04	NULL
A12345	S004	2023-08-05	NULL
A12345	S001	2023-08-06	NULL
B67890	S002	2023-08-07	NULL
C11111	S005	2023-08-08	NULL
C11111	S002	2023-06-05	NULL
A12234	NULL	NULL	NULL
D12345	NULL	NULL	NULL

```
mysql> UPDATE enrollment
-> SET date_of_enrollment =
-> CASE SID
-> WHEN 'S001' THEN '2018-01-15'
-> WHEN 'S002' THEN CURRENT_DATE
-> WHEN 'S003' THEN '2018-01-02'
-> END
-> WHERE SID IN ('S001', 'S002', 'S003');
Query OK, 7 rows affected (0.01 sec)
Rows matched: 7 Changed: 7 Warnings: 0
```

```
mysql> SELECT * FROM enrollment ;
```

Roll_no	SID	date_of_enrollment	fee_paid
X12349	S001	2018-01-15	NULL
Y56781	S002	2025-04-28	NULL
Z98769	S001	2018-01-15	NULL
X99939	S003	2018-01-02	NULL
A12345	S004	2023-08-05	NULL
A12345	S001	2018-01-15	NULL
B67890	S002	2025-04-28	NULL
C11111	S005	2023-08-08	NULL
C11111	S002	2025-04-28	NULL
A12234	NULL	NULL	NULL
D12345	NULL	NULL	NULL

```
mysql> CREATE VIEW society_enrollment_view AS
-> SELECT so.soc_name, COUNT(e.Roll_no) AS total_enrolled
-> FROM society so
-> LEFT JOIN enrollment e ON so.soc_id = e.SID
-> GROUP BY so.soc_name;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select * from society_enrollment_view;
```

soc_name	total_enrolled
NSS	3
Robotics Club	3
Drama Club	1
Science Society	1
Art Society	1

```
mysql> select s.Student_name from student s
-> join enrollment e on e.Roll_no = s.Roll_no
-> group by s.Roll_no
-> having count(distinct e.SID) = (select count(*) from society);
Empty set (0.01 sec)
```

```
mysql> SELECT COUNT(*)
-> FROM (
->     SELECT SID
->     FROM enrollment
->     GROUP BY SID
->     HAVING COUNT(Roll_no) > 2
-> ) AS temp;
```

```
+-----+
| COUNT(*) |
+-----+
|         2 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> ALTER TABLE student
-> ADD COLUMN mobile_number VARCHAR(10) DEFAULT '9999999999';
```

Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

```
mysql> select * from student ;
```

Roll_no	Student_name	Course	DOB	mobile_number
A12234	Aditi	comp sci	2003-02-28	9999999999
A12345	Annie	comp sci	2003-03-30	9999999999
B67890	Eve	biology	2002-11-25	9999999999
C11111	Frank	chemistry	2001-07-17	9999999999
D12345	Shreya	biology	2003-03-29	9999999999
X12349	Alice	comp sci	2003-05-21	9999999999
X99939	David	maths	2004-01-09	9999999999
Y56781	Bob	chemistry	2002-08-15	9999999999
Z98769	Charlie	physics	2001-12-10	9999999999

9 rows in set (0.00 sec)

```
mysql> SELECT COUNT(*)
-> FROM (
->     SELECT so.soc_id
->     FROM society so
->     JOIN enrollment e ON so.soc_id = e.SID
->     WHERE so.soc_name LIKE 'S%'
->     GROUP BY so.soc_id
->     HAVING COUNT(e.Roll_no) >= 5
-> ) AS temp;
```

```
+-----+
| COUNT(*) |
+-----+
|         0 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> SELECT so.soc_name, so.mentor_name, so.totalseats, COUNT(e.Roll_no) AS total_enrolled,
->      (so.totalseats - COUNT(e.Roll_no)) AS vacant_seats
-> FROM society so
-> LEFT JOIN enrollment e ON so.soc_id = e.SID
-> GROUP BY so.soc_name, so.mentor_name, so.totalseats
-> ;
```

soc_name	mentor_name	totalseats	total_enrolled	vacant_seats
NSS	Dr.Smith	55	3	52
Robotics Club	Ms.Johnson	33	3	30
Drama Club	Mr.Lee	44	1	43
Science Society	Dr.Watson	66	1	65
Art Society	Mrs.ADITI	28	1	27

5 rows in set (0.00 sec)